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colab.research.google.com/drive/15m8ZK5QGv_XJEvO6IARBHse5p46z-h6Q#scrollTo=iMmrXGk32EHs

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Tarea5_MAT_regresion_lineal.ipynb

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[1] 11s

```
# algebra lineal
import numpy as np

# DataFrames
import pandas as pd

# Graficar
import matplotlib.pyplot as plt

# Statsmodels (análisis estadístico)
import statsmodels.formula.api as smf
import statsmodels.api as sm

# Scikit-learn (machine learning)
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score, root_mean_squared_error
from sklearn.model_selection import train_test_split
```

Regresión lineal (desde diferentes puntos de vista)

- Python (statsmodels (`estadística`), scikit-learn(`machine learning``)
- Solución analítica (matrices)
- Solución aproximada (descenso del gradiente)

[ ]

```
url = "https://raw.githubusercontent.com/IvTole/Matematicas_Aplicadas_Ciencia_De_Datos_CUGDL/refs/heads/main/data/cars/mpg.csv"
df = pd.read_csv(url)
df = df.dropna()
df.head()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320

Variables Terminal

9:32 AM Python 3

- Regresión Lineal

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots \\ 1 & x_{21} & x_{22} & \cdots \\ \vdots & x_{31} & x_{32} & \cdots \\ 1 & & \cdots & x_{nm} \end{pmatrix}; \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}$$

> Modelo

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_m x_m$$
$$= X\beta$$

> Función de Costo

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 = \frac{1}{2n}\sum_{i=1}^{n}\left(\beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + \cdots + \beta_n x_n^{(i)} - y^{(i)}\right)^2$$

$$\frac{\partial}{\partial \beta_j} = \frac{1}{n}\sum\left(\beta_0 + \beta_1 x_1^{(i)} + \cdots + \beta_m x_m^{(i)} - y^{(i)}\right) x_j^{(i)} = X^T(X\beta - y)$$

Por derivada en forma matricial

$$= X^T(\hat{y} - y)$$

- Regresion Logistica

> Modelo

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n = X\beta$$

$$P = \frac{1}{1+e^{-X\beta}} = \sigma(X\beta)$$

> Funcion de costo (RCE)

$$\ell = \frac{1}{n}\sum_{i=1}^{n} y_i \ln(P_i) + (1-y_i)\ln(1-P_i)$$

$$\frac{\partial \ell}{\partial \beta_j} = X^T(y - P)$$

$$X^T(y - \sigma(X\beta))$$

$$\frac{\partial \ell^2}{\partial \beta_j \partial \beta_k} = X^T \Omega X$$

$$\Omega = \text{diag}(P_i(1-P_i))$$